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Experience

Hanuut Express [<https://hanuut.com>]

Co-Founder and Lead Developer

2022 – 2024

- Co-founded a startup and led technical development of an e-commerce platform enabling local shops to establish online presence and sell products digitally.
- Developed a cross-platform mobile application using Flutter for customer and shop management.
- Built backend API with NestJS and MongoDB, implementing JWT authentication/authorization and database operations for the platform.
- Built a real-time delivery system using Flutter with Google Maps integration, WebSocket connections for live route updates, and Firebase push notifications to alert nearby drivers of new orders.
- Gained valuable experience in team dynamics and meeting project deadlines.

Internships

GlaxoSmithKline(GSK)

Master's Thesis - Analyzing Deviations in Quality Control at GlaxoSmithKline (GSK)

Fev – Jul 2025

- Developed an AI-powered web application to automate root cause identification for out-of-specification (OOS) results in pharmaceutical quality control, reducing investigation time and costs.
- Built full-stack solution using Next.js frontend, Firebase for authentication and database, and Flask backend server for API management.
- Implemented a neural network using TensorFlow to solve multiclass root cause classification, achieving 98% accuracy in predictions, with scikit-learn and pandas for data preprocessing.
- Created centralized platform enabling investigation managers, laboratory staff, and production teams to efficiently exchange data and automatically distribute corrective actions (CAPAs).
- Gained valuable insights into large-scale pharmaceutical operations and enterprise-level quality control processes.
- Delivered a solution with potential to significantly reduce GSK's expensive and time-intensive investigation procedures through automated root cause identification.

Projects

Anzar Authentication Service [[Documentation](#)]

Open Source Project

2024 – Present

- Architected a production-ready authentication microservice in Rust, implementing dual JWT/Session strategies and MongoDB/SQLite support.
- Engineered a comprehensive security layer preventing CSRF, XSS, SQL injection, and DoS attacks through rate limiting, constant-time execution, and device cookies.
- Developed SDKs for Python/Ts, containerized the service as Docker image to simplify its usage, and created documentation to facilitate integration into existing applications.

Education

Master's Degree in Bio-informatics

University of Science and Technology Houari Boumediene (USTHB)

2023 – 2025

Bachelor's Degree (Licence) in Computer Systems (SI)

University of Batna 2

2020 – 2023

Technical skills

Category	Skills
Operating Systems	Linux
Version Control	Git
Databases	MongoDB
Programming Languages	TypeScript, Python, Rust (learning)
Containerization	Docker (learning)
Frameworks	Svelte, Flutter, Nestjs, Actix (learning)
Cloud Platforms	Firebase
Machine Learning	Sckit-learn, TensorFlow (learning)
